



FPGA QUICK START GUIDE

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1 LOGIN DETAILS

Console Access:

Username: `debian`

Password: `temppwd`

Baud rate: `115200 8n1`

`sudo` is enabled without password for root access

root login is disabled

2 OBTAINING THE USD IMAGE FILES

The uSD images may be downloaded via the following URLs:

Manufacturer	Board	uSD Image Download URL
Altera	A10-SoCDK	http://ndi.link/NDIUSDA10
Altera	Agilex7-SoCDK	http://ndi.link/NDIUSDAGX7
Digilent	Zybo-Z7-20	http://ndi.link/NDIUSDZYBO
Digilent	Arty-Z7-20	http://ndi.link/NDIUSDZYBO
Xilinx	ZCU104	http://ndi.link/NDIUSDZCU
Xilinx	Kria KV260	http://ndi.link/NDIUSDKV260

NOTE: The same uSD image is used for the Zybo-Z7-20 board, and the Arty-Z7-20. The uSD image contains boot files for both systems. To use the image with the Arty-Z7-20, you must update the boot files on the FAT partition prior to booting on the Arty development board.

Each uSD image is a zip archive of three files:

Extension	Description
<code>.img.xz</code>	The image file (xz compressed)
<code>.bmap</code>	A block map file for use with bmaptools (highly recommended)
<code>.md5</code>	Checksum of the above two files

Most users will likely only need to use the `img.xz` file.

3 WRITING THE IMAGE TO A USD CARD

Use of `bmaptool` at a Linux command line is the most efficient way to write the image file to a uSD card. If this is not an option or is considered too complex, any uSD imaging tool which supports xz compression can be used. [Etcher](#) is a good choice if you do not already have a preference.

4 USING THE IMAGE

Insert the programmed uSD card into the development board and boot as usual. It is recommended you connect to the serial terminal (via USB). All images use the following console settings:

```
115200 baud, 8 data bits, 1 stop bit, no parity
```

The system will boot and display the IP address, as well as the default username and password at the serial console login prompt. You may login either via the serial console or remotely via ssh.

4.1 FIRST BOOT

The initial boot will take somewhat longer than normal as a new set of ssh keys are generated and the root partition is expanded to fill the uSD card.

5 RUNNING THE EXAMPLE NDI ENCODE APPLICATION

An example NDI source application is provided. This application is launched automatically at boot and may also be run from the command line. The utility is named `ndi_encode` and is capable of using a generated video pattern or live HDMI input as the video source. The application includes various command-line options to control the initial video mode as well as a command interpreter that can be used to change the operating mode at run-time. To launch the demo from the command line, simply run one of the following commands:

```
# Live HDMI input:
ndi_encode

# Pattern generator:
ndi_encode -P
```

Once running, the operating mode can be changed by typing commands at the console. Type “help” to get a list of commands.

If running with HDMI input on the ZCU104, the second virtual serial port is used as a console for the Cortex-R5 running the HDMI monitoring software. Connection details are the same as the Linux console:

```
115200 baud, 8 data bits, 1 stop bit, no parity
```

The R5 will report changes in the incoming HDMI stream on this port.

5.1 KNOWN ISSUES AND LIMITATIONS

While the NDI Encoder core is fully functional the example software and demo platforms currently have some limitations:

- This version of the NDI Advanced SDK is designed for development use and will run on a stream for 30 minutes. For a commercial use license, please email support@ndi.video
- Due to the asynchronous nature of the startup scripts, the system clock may be adjusted after the `ndi_encode` example is launched. If this happens, it may trigger the 30 minute timeout even though the application may not have been running for 30 minutes of “wall clock” time. Kill and re-start the demo application to resume NDI streaming.
- The HDMI input hardware for the ZCU104 uses an evaluation license, so HDMI video will stop being received and the image will turn “green” after apx. 1 hour of operation. Reprogram the FPGA (eg: reboot) to restore normal operation.
- While HDMI input on the ZCU104 is functional, the HDMI management software (running on one of the R5 cores) does not communicate with the `ndi_encode` application. The video format tracking hardware is used to detect video format changes, however more details regarding the HDMI signal are available via the HDMI management software.

6 RUNNING THE EXAMPLE NDI DECODE APPLICATION

An example NDI receive application is provided. This application is launched manually by executing `ndi_decode`. When launched with no options the utility will attempt to locate NDI sources on the local network and will automatically connect to the first source found. A specific NDI source may be specified using the `-s` parameter:

```
ndi_decode -s "NDI Device (Chan 1)"
```

NOTE: The default boot files on the uSD image must be switched to the NDI Decode version prior to running the `ndi_decode` application. See the “Available Examples” section for your specific board for details.

6.1 KNOWN ISSUES AND LIMITATIONS

- “Scaling” is performed by manipulating line stride when the NDI stream and video output resolutions do not match
- Format conversion is not performed between 4:2:2 and 4:2:0 video sources. If the output video format does not match the NDI stream format, a warning is displayed on the Linux console.

7 BOARD DETAILS

Connections and settings required to boot using the uSD images. Refer to the vendor documentation for full details.

7.1 ALTERA ARRIA-10 SOC DEVELOPMENT KIT

Part number: DK-SOC-10AS066S-D

7.1.1 AVAILABLE EXAMPLES

Two FPGA examples are available for the Arria-10 SoC board:

- Decode example with 4 NDI Decode cores
- Encode example with 4 NDI Encode cores

The default uSD image uses the Enc files, however files for all examples are included on the uSD card. To change the example design, simply update the FPGA programming files in the `/boot` directory and **cycle power**:

```
# Use the Decode example
sudo cp /boot/fit_spl_fpga_dec.itb /boot/fit_spl_fpga.itb

# Use the Encode example
sudo cp /boot/fit_spl_fpga_enc.itb /boot/fit_spl_fpga.itb

# You MUST cycle power to reprogram the FPGA!!!
```

7.1.2 JUMPERS AND SWITCHES

All jumpers and switches are set per the defaults listed in section 3 of the Arria 10 SoC Development Kit User Guide.

Ref	Function	Setting
SW1	Boot Mode	All off
SW2	User Switches	All off
SW3.1	A10 JTAG	Off
SW3.2	Max V JTAG	Off
SW3.3	FMCA JTAG	On
SW3.4	FMCB JTAG	On
SW3.5	PCIe JTAG	On
SW3.6	MSTR0 JTAG	Off
SW3.7	MSTR1 JTAG	Off
SW3.8	MSTR2 JTAG	Off
SW4	MSEL	All off
J16	OSC2 Clk Sel	Short
J17	OSC2 Clk Sel	Short
J30	HPS Core Voltage	Short
J32	FMCBVADJ	Short 9 and 10
J42	FMCVADJ	Short 9 and 10
J3	BSEL0	Short left 2 pins

Ref	Function	Setting
J4	BSEL1	Short upper 2 pins
J5	BSEL2	Short upper 2 pins

7.1.3 CONNECTIONS

Ref	Function / Label	Connect to
J19	FMC Port B	Bitec FMC HDMI Daughter Card Rev. 11
J5	HPS Ethernet	Local Ethernet network
J10	PROG/UART	Host system USB port
J36	Power	Power Supply Module
RX	Bitec HDMI Rx	HDMI video source (NDI Encode)
TX	Bitec HDMI Tx	HDMI video output (NDI Decode)

7.1.4 LEDS

LED	Function
FPGA_LED0	Tally (Main or Preview)
FPGA_LED1	Tally (Main)
FPGA_LED2	Tally (Preview)
FPGA_LED3	CPU load

7.2 ALTERA AGILEX-7 I-SERIES TRANSCEIVER-SOC DEVKIT

Part number: DK-SI-AGI027FA

7.2.1 AVAILABLE EXAMPLES

One FPGA example is available for the Agilex-7 SoC board:

- Encode example with 4 NDI Encode cores

7.2.2 BOARD PREPERATION

Connect the following daughtercards to the Agilex-7 board. All daughtercards except for the Bitec HDMI 2.1 board are included with the devkit.

- QSPI Flash Card to J3 SDM_DAUGHTER_CARD
- HPS IO48 OOB Card to J4 HPS_DAUGHTER_CARD
- DDR4 DIMM to the DIMM slot furthest from the FPGA
- Bitec HDMI 2.1 Rev. 9 Daughtercard to FMC+ B

7.2.2.1 PROGRAM JIC FILE

Prior to using the uSD image, the QSPI flash **must** be programmed with a JIC file matching the project. The JIC file `Agilex7-Enc.hps.jic.gz` can be found on the FAT partition of the uSD card image. The programming process is:

- Set S9[1:4] = ON/ON/ON/X
- Connect a USB cable from your Quartus system to J10
- Power on the Agilex-7 board
- Program the QSPI flash with the commands:
 - `gunzip path/to/Agilex7-Enc.hps.jic.gz`
 - `quartus_pgm -c 1 -m jtag -o "pvi;path/to/Agilex7-Enc.hps.jic"`
- Power down the Agilex-7 board
- Set S9[1:4] = ON/OFF/OFF/X

7.2.2.2 PROGRAM CLOCK FREQUENCIES

IMPORTANT! This must be done everytime the Agilex-7 board is power cycled!

The frequencies of the Si5391-B clock synthesizer chip need to be manually configured as described in the [HDMI IP Design Example Quick Start Guide](#):

- Connect a system to J7, the USB serial console port on the HPS IO48 OOB Card
- Connect a USB cable from your Quartus system to J10
- Insert the uSD image into the HPS IO48 OOB Card uSD socket
- Power on the Agilex-7 board
- Press a key at the U-Boot prompt to prevent the system from booting into Linux
- Launch the `ClockController` utility found in the `examples/board_test_system` directory from the Installer Package for the board
- Select the tab for the Si5391-B clock chip
- Enable all outputs
- Set all output frequencies to 100.00 MHz
- Click `Set` and close the ClockController application
- Reset the board by pressing SYS_PB3 (Power recycle)

- Allow U-Boot to boot Linux normally

7.2.3 JUMPERS AND SWITCHES

Most jumpers and switches are set per the defaults listed in the User Guide. Note that S9 must be modified when programming the JIC file (see above) and then returned to it's default value for normal operation.

S23[1] = OFF is the only switch that needs to be changed from it's default value.

Ref	Setting	Function
S19[1:4]	OFF/OFF/ON/ON	System MAX® 10 and FPGA selected in the JTAG chain
S20[1:4]	ON/ON/ON/ON	Mode 1: Onboard Altera® download circuit act as the only JTAG master
S9[1:4]	ON/OFF/OFF/X	Configuration mode setting bits
S10[1:4]	ON/ON/ON/ON	SYS_SW[0:3]
S15[1:4]	ON/ON/ON/OFF	SYS_SW[4:7]
S1[1:4]	OFF/OFF/OFF/OFF	User Switch [0:3]
S6[1:4]	OFF/OFF/OFF/OFF	User Switch [4:7]
S22[1:4]	ON/ON/ON/ON	MUX_DIP_SW[0:3]
S23[1:4]	ON/ON/ON/ON	MUX_DIP_SW[4:7] NON-DEFAULT SETTING
S4[1:4]	ON/ON/ON/ON	MUX_DIP_SW[8:11]

7.2.4 CONNECTIONS

Ref	Function / Label	Connect to
FMC+ B	FMC Port B	Bitec FMC HDMI 2.1 Daughter Card Rev. 9
OUBE.J3	HPS Ethernet	Local Ethernet network
J10	PROG	Host system USB port used for programming
OUBE.J7	UART	Host system USB port used for HPS serial console
J18	Power	Power Supply Module
Bitec RX	Bitec HDMI Rx	HDMI video source (NDI Encode)
Bitec TX	Bitec HDMI Tx	HDMI video output (NDI Decode)

7.2.5 LEDS

LED	Function
FPGA_LED0	Tally (Main or Preview)
FPGA_LED1	Tally (Main)
FPGA_LED2	Tally (Preview)
FPGA_LED3	CPU load

7.3 DIGILENT ZYBO-Z7-20

Part number: 410-351-20

7.3.1 AVAILABLE EXAMPLES

Three FPGA examples are available for the Zybo-Z7 board:

- Zybo-Z7-20-Dec: 4-core NDI Decoder with 1 GB of 32-bit SDRAM
- Zybo-Z7-20-Enc: 4-core NDI Encoder with 1 GB of 32-bit SDRAM
- Zybo-Z7-20-Enc-Lite: 2-core NDI Encoder with 512 MB of 16-bit SDRAM (1 SDRAM chip is unused)

The default uSD image uses the 4-core Encoder files, however files for all three examples are included on the uSD card. To change the example design, simply update the files in the `/boot` directory and reboot:

```
# Use the Decode example with 4 cores and 1 GB of 32-bit SDRAM
sudo cp /boot/Zybo-Z7-20-Dec/* /boot/
```

```
# Use the Encode example with 4 cores and 1 GB of 32-bit SDRAM
sudo cp /boot/Zybo-Z7-20-Enc/* /boot/
```

```
# Use the "Lite" Encode example with 2 cores and 512 MB of 16-bit SDRAM
sudo cp /boot/Zybo-Z7-20-Enc-Lite/* /boot/
```

The “Enc-Lite” Encode example is intended to allow performance evaluation of a typical “minimal” Zynq platform that uses only a single SDRAM memory chip with a 16-bit memory bus. This design is still capable of processing 1080p60 video.

The “Full” Encode example will have lower latency at all resolutions and can in theory compress up to 4Kp60 4:2:0 video (with enough available SDRAM bandwidth). A source of 4K video would also be required (eg: Analog Devices ADV7619, ITE IT68059) since the HDMI I/O on the Zybo-Z7 does not support resolutions beyond 1080p60.

7.3.2 JUMPERS AND SWITCHES

Ref	Function	Setting
JP5	Boot Mode	SD
JP6	Power	WALL

7.3.3 CONNECTIONS

Ref	Function / Label	Connect to
J9	HDMI RX	HDMI video source (NDI Encode)
J8	HDMI TX	HDMI video output (NDI Decode)
J7	LINE IN	Analog audio in
J5	HPH OUT	Analog audio out
J3	ETH	Local Ethernet network
J17	Power	Appropriate 5V power supply
J12	PROG/UART	Host system USB port

7.3.4 LEDS

LED	Function
LD0	Tally (Main)
LD1	Tally (Preview)
LD2	CPU load
LD3	uSD activity
LD4	Tally (either Main or Preview)

7.4 DIGILENT ARTY-Z7-20

Part number: 410-346-10

7.4.1 AVAILABLE EXAMPLES

One FPGA example is available for the Arty-Z7 board:

- Arty-Enc.Xil: NDI Encoder based on Zybo “Lite” design

The default uSD image uses the Zybo boot files, however the Arty boot files are included. To modify the uSD image to boot on the Arty, simply update the files in the FAT partition before booting the Arty board:

```
# Mount the first partition on the uSD
sudo mount /dev/sdx1 /mnt

# Copy the Arty version of the "Lite" Encode example to the filesystem root
sudo cp /mnt/Arty-Z7-20-Enc/* /mnt/

# Unmount the partition
sudo umount /mnt

# Remove the uSD and use to boot the Arty-Z7
```

7.4.2 JUMPERS AND SWITCHES

Ref	Function	Setting
JP4	Boot Mode	SD
JP5	Power	USB (when using USB power)
JP5	Power	REG (when using J18 with 7-15V power)

7.4.3 CONNECTIONS

Ref	Function / Label	Connect to
J10	HDMI IN	HDMI video source (NDI Encode)
J11	HDMI OUT	HDMI video output (NDI Decode)
J8	ETHERNET	Local Ethernet network
J14	PROG/UART	Host system USB port

7.4.4 LEDS

LED	Function
LD0	Tally (Main)
LD1	Tally (Preview)
LD2	CPU load
LD3	uSD activity

7.5 XILINX ZCU104

Part number: EK-U1-ZCU104-G

7.5.1 AVAILABLE EXAMPLES

Two FPGA examples are available for the ZCU104 board:

- Dec: Decode example with 4 NDI Decode cores
- Enc: Encode example with 4 NDI Encode cores

The default uSD image uses the Enc files, however files for all examples are included on the uSD card. To change the example design, simply update the files in the `/boot` directory and reboot:

```
# Use the Decode example
sudo cp /boot/ZCU104-Dec/* /boot/
```

```
# Use the Encode example
sudo cp /boot/ZCU104-Enc/* /boot/
```

7.5.2 JUMPERS AND SWITCHES

Ref	Function	Setting
J85	POR_OVERRIDE	2-3 (default)
J12	SYSMON I2C addr	1-2 (default)
J13	SYSMON I2C addr	1-2 (default)
J20	PS_POR_B	1-2 (default)
J21	PS_SRST_B	1-2 (default)
J22	Reset Seq	open (default)
SW6 1	PS_MODE	On
SW6 2	PS_MODE	Off
SW6 3	PS_MODE	Off
SW6 4	PS_MODE	Off

7.5.3 CONNECTIONS

Ref	Function / Label	Connect to
P7	HDMI Rx (bottom)	HDMI video source
J52	Power	Power supply
P12	Ethernet	Ethernet network
J164	JTAG/UART	Host system USB port

7.5.4 LEDS

LED	Function
LED0	Tally (Main or Preview)
LED1	Tally (Main)
LED2	Tally (Preview)
LED3	CPU load

7.6 XILINX KRIA KV260

Part number: SK-KV260-G

7.6.1 AVAILABLE EXAMPLES

One FPGA example is available for the Kria KV260:

- Enc: Encode example with 4 NDI Encode cores

7.6.2 BOARD SETUP

The NDI example design requires the same board configuration as the Xilinx “Smart Camera” example, including the FSBL programmed into the on-module flash memory and the AR1335 camera module as provided in the [Basic Accessory Pack](#) connected to J7.

See the [Getting Started](#) instructions from Xilinx for full details.

There are no jumpers or switches to set on the KV260, and there are no user LEDs available to indicate tally.